

The Central Measure of the Unit Interval Is Lebesgue Measure

Run fa_banach_001, lane 13

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Abstract

Niemiec defined the central measure of a compact metric space as the generalized Chebyshev center of its probability measures in the Kantorovich metric, and asked whether the central measure of $[0, 1]$ is Lebesgue measure. We give an affirmative answer. The first Chebyshev center consists of the measures of mean $1/2$. In quantile coordinates this center has a nonlinear involutive isometry that exchanges a monotone function with its distribution function. Its only fixed point is the identity function. Since the generalized Chebyshev center is fixed by every isometry, its quantile is the identity, hence the measure is Lebesgue.

1 The question and the answer

Let $I = [0, 1]$. Write $\mathcal{P}(I)$ for the Borel probability measures on I , equipped with the Kantorovich–Rubinstein metric

$$W_1(\mu, \nu) = \sup_{\text{Lip}(f) \leq 1} \left| \int f d\mu - \int f d\nu \right|.$$

For a compact convex subset K of a normed space, let $C(K)$ denote its Chebyshev center set, put $C^0(K) = K$ and $C^{n+1}(K) = C(C^n(K))$, and let c_K be the unique member of $\bigcap_{n \geq 0} C^n(K)$. The source calls $c_{\mathcal{P}(I)}$ the *central measure* μ_I .

On PDF page 7, after observing that the isometry group of the interval gives only reflection invariance, the source asks whether μ_I is Lebesgue measure; see Figure 1.

Theorem 1 (Full answer). *The central measure $\mu_{[0,1]}$ is Lebesgue measure on $[0, 1]$.*

Corollary 4.3 provides a new constructive proof of the Haar measure theorem for metrizable compact groups.

Although in compact metric spaces with transitive actions of the isometry groups the central measures may be found thanks of their very specific properties, unfortunately computing a central measure in general is very complicated. For example, we do not know the one of $[0, 1]$. The reader interested in this problem may try first to compute the central measure of a three-point space.

Question. Is $\mu_{[0,1]}$ the Lebesgue measure?

Figure 1: The source question, arXiv:1105.5706, PDF p. 7.

2 Idea of the proof

The reflection $x \mapsto 1 - x$ fixes every symmetric probability measure, so it cannot identify the central measure by itself. The useful symmetry appears only after taking the first Chebyshev center.

Represent a measure by its quantile q . The first center is precisely

$$\mathcal{Q}_{1/2} = \left\{ q : q \text{ is nondecreasing and } \int_0^1 q(u) du = \frac{1}{2} \right\}.$$

For $q \in \mathcal{Q}_{1/2}$ set

$$(\mathcal{D}q)(t) = \mathbf{m}\{u \in [0, 1] : q(u) \leq t\}.$$

Geometrically, $\mathcal{D}$ reflects the completed graph of q across the diagonal. Areas of symmetric differences show that it preserves L^1 distance, and the area below the graph shows that it sends $\int q$ to $1 - \int q$. Consequently $\mathcal{D}$ is an isometry of $\mathcal{Q}_{1/2}$.

The generalized Chebyshev center is fixed by every isometry of its compact convex metric space. Thus its quantile is fixed by $\mathcal{D}$. A monotonicity argument shows that $\mathcal{D}q = q$ forces $q(t) = t$. This is exactly the quantile of Lebesgue measure.

3 The first Chebyshev center

For $\mu \in \mathcal{P}(I)$ write

$$m(\mu) = \int_I x d\mu(x), \quad e(\mu) = \sup_{\nu \in \mathcal{P}(I)} W_1(\mu, \nu).$$

Lemma 2. *For every $\mu \in \mathcal{P}(I)$,*

$$e(\mu) = \max\{m(\mu), 1 - m(\mu)\}.$$

Consequently the Chebyshev radius of $\mathcal{P}(I)$ is $1/2$ and

$$C(\mathcal{P}(I)) = \{\mu \in \mathcal{P}(I) : m(\mu) = 1/2\}.$$

Proof. For any ν , the independent coupling gives

$$W_1(\mu, \nu) \leq \int_I \int_I |x - y| d\mu(x) d\nu(y) \leq \max_{0 \leq y \leq 1} \int_I |x - y| d\mu(x).$$

The last function is convex in y , so its maximum is attained at an endpoint. Its value is

$$\max\{m(\mu), 1 - m(\mu)\}.$$

Conversely,

$$W_1(\mu, \delta_0) = m(\mu), \quad W_1(\mu, \delta_1) = 1 - m(\mu).$$

This proves the formula, whose minimum is $1/2$ exactly when $m(\mu) = 1/2$. □

4 Quantile coordinates and the hidden isometry

Let $Q_\mu : (0, 1) \rightarrow I$ be the usual nondecreasing quantile function of μ . Let $\mathcal{Q}$ be the set of equivalence classes modulo null sets of nondecreasing measurable maps $q : I \rightarrow I$.

Lemma 3 (One-dimensional Kantorovich coordinates). *The map $\mu \mapsto Q_\mu$ is a bijective isometry*

$$(\mathcal{P}(I), W_1) \longrightarrow (\mathcal{Q}, \|\cdot\|_{L^1}), \quad W_1(\mu, \nu) = \int_0^1 |Q_\mu(u) - Q_\nu(u)| du.$$

Moreover,

$$m(\mu) = \int_0^1 Q_\mu(u) du.$$

Proof. The second identity follows by taking U uniform on I : $Q_\mu(U)$ has law μ . For completeness, if F_μ, F_ν are the distribution functions, one-dimensional Kantorovich duality and integration by parts give

$$W_1(\mu, \nu) = \int_0^1 |F_\mu(t) - F_\nu(t)| dt.$$

The region between two monotone completed graphs has the same area whether integrated horizontally or vertically. Fubini's theorem therefore gives

$$\int_0^1 |F_\mu - F_\nu| = \int_0^1 |Q_\mu - Q_\nu|.$$

Every nondecreasing $q : I \rightarrow I$ is, modulo a null set, the quantile of the law of $q(U)$, which proves surjectivity. $\square$

Under this isometry, Lemma 2 identifies the first center with the compact convex set

$$\mathcal{Q}_{1/2} = \left\{ q \in \mathcal{Q} : \int_0^1 q = \frac{1}{2} \right\}.$$

Lemma 4 (Inverse-distribution isometry). *For $q \in \mathcal{Q}$ define, at every $t \in I$,*

$$(\mathcal{D}q)(t) = \mathfrak{m}\{u \in I : q(u) \leq t\}.$$

Then $\mathcal{D} : \mathcal{Q} \rightarrow \mathcal{Q}$ is an involutive isometry modulo null sets, and

$$\int_0^1 \mathcal{D}q(t) dt = 1 - \int_0^1 q(u) du.$$

In particular, $\mathcal{D}$ is a surjective isometry of $\mathcal{Q}_{1/2}$ onto itself.

Proof. For fixed t , the sublevel set $A_t(q) = \{u : q(u) \leq t\}$ is, modulo endpoints, an initial interval. Hence

$$|\mathcal{D}q(t) - \mathcal{D}r(t)| = \mathfrak{m}(A_t(q) \triangle A_t(r)).$$

Integrating this identity and using Fubini gives

$$\begin{aligned} \|\mathcal{D}q - \mathcal{D}r\|_1 &= \int_0^1 \int_0^1 \left| \mathbf{1}_{\{q(u) \leq t\}} - \mathbf{1}_{\{r(u) \leq t\}} \right| du dt \\ &= \int_0^1 |q(u) - r(u)| du. \end{aligned}$$

Similarly,

$$\int_0^1 \mathcal{D}q(t) dt = \int_0^1 \int_0^1 \mathbf{1}_{\{q(u) \leq t\}} dt du = \int_0^1 (1 - q(u)) du.$$

Finally, reflecting a monotone completed graph twice recovers it; equivalently, the elementary generalized-inverse identity gives $\mathcal{D}^2 q = q$ almost everywhere. Thus $\mathcal{D}$ is involutive and hence surjective. $\square$

5 The fixed point of the hidden isometry

Lemma 5. *The only $q \in \mathcal{Q}$ satisfying $\mathcal{D}q = q$ almost everywhere is $q(t) = t$ almost everywhere.*

Proof. Put $F = \mathcal{D}q$. If $q = F$ almost everywhere, then sublevel-set measures do not change when q is replaced by F . Therefore

$$\mathcal{D}F = \mathcal{D}q = F$$

pointwise. We show that a nondecreasing $F : I \rightarrow I$ with $\mathcal{D}F = F$ must be the identity.

Suppose first that $F(t) > t$ for some t . Choose $0 < \varepsilon < F(t) - t$. For every $s \in (t, t + \varepsilon)$ with $s < F(t)$, monotonicity implies

$$\{x : F(x) \leq s\} \subset [0, t), \quad \text{so} \quad (\mathcal{D}F)(s) \leq t,$$

whereas $F(s) \geq F(t) > t$. This contradicts $\mathcal{D}F = F$.

If $F(t) < t$, choose $0 < \varepsilon < t - F(t)$. For every $s \in (t - \varepsilon, t)$ with $s > F(t)$, monotonicity gives

$$[0, t] \subset \{x : F(x) \leq s\}, \quad \text{so} \quad (\mathcal{D}F)(s) \geq t,$$

while $F(s) \leq F(t) < t$, again a contradiction. Hence $F(t) = t$ for every t , and therefore $q(t) = t$ almost everywhere. $\square$

6 Completion of the proof

Let $c_{\mathcal{Q}}$ be the generalized Chebyshev center of $\mathcal{Q}$. Since $C(\mathcal{Q}) = \mathcal{Q}_{1/2}$, the iterates satisfy

$$C^n(\mathcal{Q}_{1/2}) = C^{n+1}(\mathcal{Q}) \quad (n \geq 0),$$

and therefore

$$c_{\mathcal{Q}} = c_{\mathcal{Q}_{1/2}}.$$

The source proves that the generalized Chebyshev center of every nonempty compact convex subset of a normed space is fixed by every bijective isometry of that metric space. Lemma 4 gives such an isometry $\mathcal{D}$ of $\mathcal{Q}_{1/2}$. Hence

$$\mathcal{D}(c_{\mathcal{Q}}) = c_{\mathcal{Q}}.$$

Lemma 5 now yields $c_{\mathcal{Q}}(t) = t$ almost everywhere. The measure whose quantile is $t \mapsto t$ is Lebesgue measure on I . Transporting back through Lemma 3 proves Theorem 1.

7 Checks, scope, and novelty

The proof concerns the source’s central measure of the *first kind*, namely the generalized Chebyshev center of $\mathcal{P}(I)$. It does not substitute the source’s separately defined central measure of the second kind, which the paper already proves to be Lebesgue.

The choice of $\leq$ in the definition of $\mathcal{D}$ affects values only at the countably many discontinuity levels of a monotone function. All metric and involution statements are in L^1 , hence modulo null sets. In the fixed-point argument we pass to the canonical pointwise representative $F = \mathcal{D}q$, for which $\mathcal{D}F = F$ holds pointwise.

The run’s cheap indexes, the local source-probed corpus, and bounded exact-phrase/title searches on July 21, 2026 found the original paper and self-citations but no later answer to this exact question. This is therefore packaged as a candidate full solution with moderate, not definitive, novelty confidence.

Reference

P. Niemiec, *Central points and measures and dense subsets of compact metric spaces*, Topological Methods in Nonlinear Analysis 40 (2012), 161–180; arXiv:1105.5706.